

SEID 2364 – Assignment 6 Mediation Pathways Across Physical and Virtual Space

Part 1 – System Restatement

Decision Statement

An AI hiring system evaluates and filters job applicants for employment opportunities.

Actors

The main actors are job applicants, recruiters, company management, software developers, and the AI system itself.

Targets

The primary targets are applicants whose resumes are evaluated. Secondary targets include families and communities affected by employment outcomes.

Determinants and Contributors

Determinants include hiring criteria, training data, company policies, and machine learning algorithms. Contributors include HR staff, historical hiring records, and third-party AI vendors.

Key Data Inputs

The system uses resumes, educational records, work history, skills, and assessment scores.

Part 2 – Mediation Pathway Inventory

Data Pathway

Source → Applicant Resume → AI Processing System → Recruiter Dashboard

The pathway begins with the applicant submitting personal and professional information. The AI system processes this information using machine learning models and sends ranked results to recruiters.

Physical Agency: Applicants provide information physically through computers or devices, while recruiters act within company hiring departments.

Virtual Agency: The AI system processes and transforms information through software, databases, and algorithms.

Information Access: Applicants can view their own submissions, while recruiters and company systems can view rankings and recommendations.

Decision Pathway

Source → AI Recommendation → Recruiter Review → Hiring Decision

The AI system generates recommendations based on applicant data. Recruiters then review these outputs before making final decisions.

Physical Agency: Recruiters make employment decisions in organizational settings.

Virtual Agency: Algorithms influence candidate rankings and recommendations.

Accountability Pathway

Source → Company Policies → HR Department → Applicant Outcome

The company establishes hiring policies and deploys the AI system. HR departments apply those policies, and applicants experience the results of the system.

Part 3 – Relativistic Physical and Virtual Travel Description

The point of origin for this analysis is the job applicant because the applicant experiences the direct consequences of the hiring process.

In physical space, applicants submit resumes from personal devices while recruiters operate within company offices. Data travels through servers and databases before reaching recruiters.

In virtual space, resumes move through algorithms, machine learning systems, and ranking models. Information is transformed from raw applicant data into recommendation scores.

The applicant may not fully observe how the system processes data, while recruiters and developers have greater visibility into internal operations. This creates asymmetry in observability and authority.

Part 4 – Pathway Transformation and Friction Analysis

One major transformation occurs when resumes are converted into ranking scores by machine learning models. Important information may be simplified or ignored during this process.

Another friction point occurs when recruiters rely heavily on AI recommendations without fully understanding how decisions were generated. This may increase opacity and reduce accountability.

Information access is also uneven. Developers and companies can observe system performance data, while applicants may only see final outcomes without explanation.

Authority and responsibility may diverge because companies deploy the system, developers build it, and recruiters apply it, but applicants experience the harm directly.

Conclusion

This assignment showed that AI hiring systems operate through multiple mediation pathways connecting physical and virtual spaces. Decisions are distributed across technical systems, institutions, and human actors. Ethical concerns increase when authority, visibility, and responsibility become unevenly distributed across the system.